IN241 Statistika

Lecture 13

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The Analysis of Variance (ANOVA) [Single Factor]

***Based on textbooks of [Ross , 2004] and [Devore, 2012].**

The Analysis of Variance

The *analysis of variance* (ANOVA), refers to a collection of experimental situations and statistical procedures for the analysis of quantitative responses from experimental units.

Terminology

The characteristic that differentiates the treatments or populations from one another is called the *factor* under study, and the different treatments or populations are referred to as the *levels* of the factor.

Single-Factor ANOVA

Single-factor ANOVA focuses on a comparison of more than two population or treatment means. Let

$I =$ the number of treatments (populations) being compared.

$\mu_1 =$ the mean of population 1 or the true average response when treatment 1 is applied

$\vdots$

$\mu_I =$ the mean of population I or the true average response when treatment I is applied

Then the hypotheses of interest are

$$H_0: \quad \mu_1 = \mu_2 = \ldots = \mu_I$$

versus

H_a: at least two of the μ_i's are different

Notation

$X_{i,j}$ = The random variable that denotes the *j*th measurement taken from the *i*th population, or the measurement taken on the *j*th experimental unit that receives the *i*th treatment

$x_{i,j}$ = The observed value of $X_{i,j}$ when the experiment is performed

Assumptions

The I population or treatment distributions are all normal with the same variance σ^2. Each $X_{i,j}$ is normally distributed with

$$E(X_{i,j}) = \mu_i \quad V(X_{i,j}) = \sigma^2$$

Notation

$$\bar{X}_{i.} = \frac{\sum_{j=1}^{J} X_{ij}}{J}$$ (Average when factor *A* is held at level *i*)

$$\bar{X}_{.j} = \frac{\sum_{i=1}^{I} X_{ij}}{I}$$ (Average when factor *B* is held at level *j*)

$$\bar{X}_{..} = \frac{\sum_{i=1}^{I}\sum_{j=1}^{J} X_{ij}}{IJ}$$ (The Grand Mean)

$$S_i^2 = \frac{1}{J-1}\sum_{j=1}^{J}(X_{ij} - \bar{X}_{i.})^2,\ i = 1 \ldots I$$ (Variances)

The Mean Square for Treatments and Error

Mean square for treatments:

$$\mathrm{MSTr} = \frac{J}{I-1}[(\bar{X}_{1\cdot} - \bar{X}_{\cdot\cdot})^2 + \ldots + (\bar{X}_{I\cdot} - \bar{X}_{\cdot\cdot})^2]$$

$$= \frac{J}{I-1}\sum_i (\bar{X}_{i\cdot} - \bar{X}_{\cdot\cdot})^2$$

Mean square for error:

$$\mathrm{MSE} = \frac{S_1^2 + S_2^2 + \ldots + S_I^2}{I}$$

The Test Statistic

The test statistic for single-factor ANOVA is

$$\boxed{f = \frac{\text{MSTr}}{\text{MSE}}}$$

Expected Value

When H_0 is true,

$$E(\text{MSTr}) \ = \ E(\text{MSE}) = \sigma^2$$

When H_0 is false,

$$E(\text{MSTr}) \ > \ E(\text{MSE}) = \sigma^2$$

F Distributions and Test

Let F = MSTr/MSE be the statistic in a single-factor ANOVA problem involving I populations or treatments with a random sample of J observations from each one. When H_0 is true (basic assumptions true) , F has an F distribution with $v_1 = I - 1$ and $v_2 = I(J - 1)$. The rejection region $f \geq F_{\alpha, I-1, I(J-1)}$ specifies a test with significance level α.

Formulas for ANOVA

Total sum of squares (SST)

$$\mathrm{SST} = \sum_{i=1}^{I}\sum_{j=1}^{J}(x_{ij} - \bar{x}_{..})^2 = \sum_{i=1}^{I}\sum_{j=1}^{J}x_{ij}^2 - \frac{1}{IJ}x_{..}^2$$

Treatment sum of squares (SSTr)

$$\mathrm{SSTr} = \sum_{i=1}^{I}\sum_{j=1}^{J}(\bar{x}_{i.} - \bar{x}_{..})^2 = \frac{1}{J}\sum_{i=1}^{I}x_{i.}^2 - \frac{1}{IJ}x_{..}^2$$

Error sum of squares (SSE)

$$\mathrm{SSE} = \sum_{i=1}^{I}\sum_{j=1}^{J}(x_{ij} - \bar{x}_{i.})^2$$

Formulas for ANOVA

Mean square for treatments:

$$\text{MSTr} = \frac{J}{I-1}\sum_{i=1}^{I}(\bar{x}_{i.} - \bar{x}_{..})^2$$

Mean square for error:

$$\text{MSE} = \frac{1}{I}\sum_{i=1}^{I} s_i^2$$

Formulas for ANOVA

where

$$\bar{x}_{i.} = \frac{1}{J}\sum_{j=1}^{J} x_{ij} \quad , \; i = 1 \ldots I$$

$$x_{i.} = \sum_{j=1}^{J} x_{ij} \quad , \; i = 1 \ldots I$$

$$\bar{x}_{..} = \frac{1}{IJ}\sum_{i=1}^{I}\sum_{j=1}^{J} x_{ij}$$ (grand mean)

$$x_{..} = \sum_{i=1}^{I}\sum_{j=1}^{J} x_{ij}$$ (grand sum)

Fundamental Indentity

$$\boxed{\mathrm{SST} = \mathrm{SSTr} + \mathrm{SSE}}$$

$$\boxed{\mathrm{MSTr} = \frac{\mathrm{SSTr}}{I-1}} \qquad \boxed{\mathrm{MSE} = \frac{\mathrm{SSE}}{I(J-1)}}$$

$$\boxed{f = \frac{\mathrm{MSTr}}{\mathrm{MSE}}}$$

The Test at Significance Level α

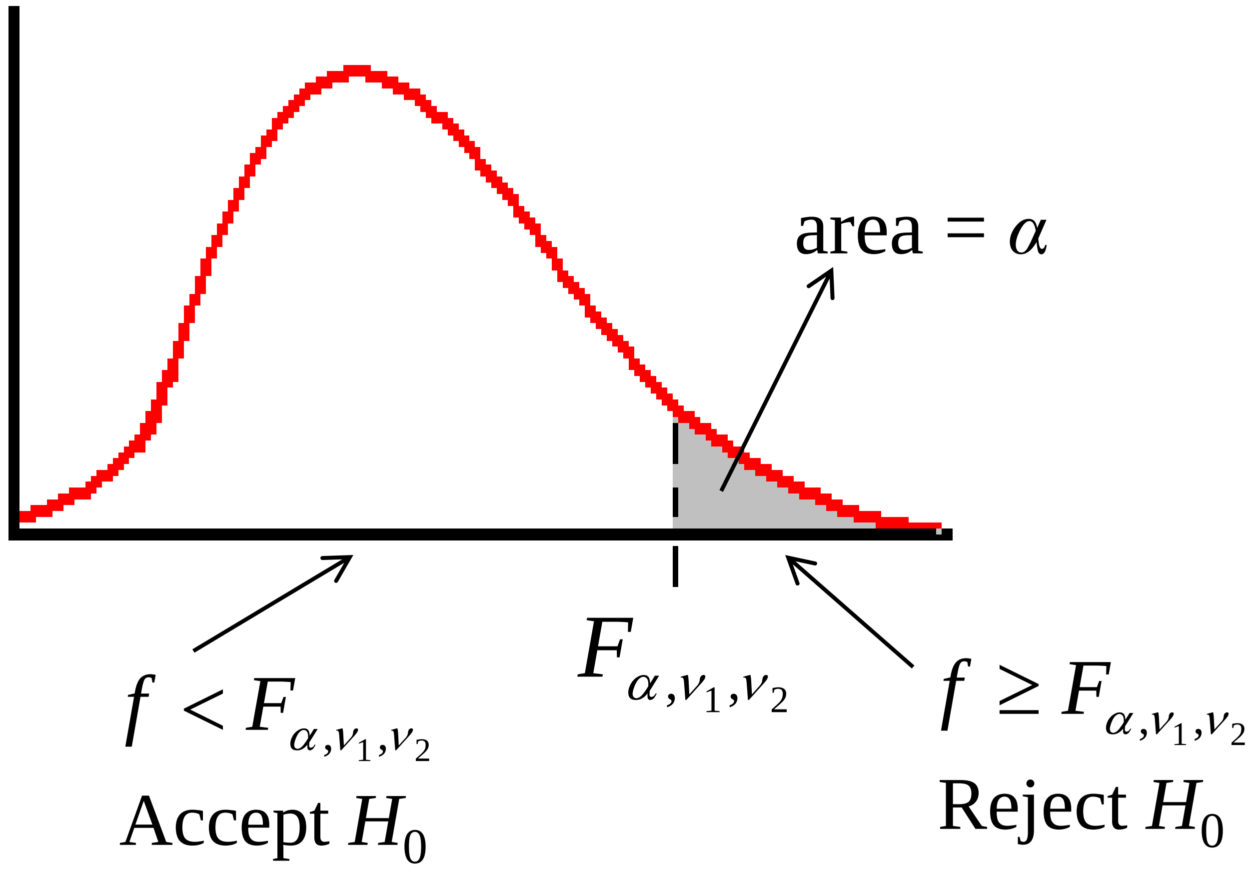
$$\boxed{f = \frac{\text{MSTr}}{\text{MSE}}}$$

$f < F_{\alpha,\nu_1,\nu_2}$ → Accept H_0 (fail to reject H_0)

$f \geq F_{\alpha,\nu_1,\nu_2}$ → Reject H_0

df's: $\nu_1 = I - 1, \quad \nu_2 = I(J-1)$

The Test at Significance Level α With F distribution



df's: $\nu_1 = I - 1 \qquad \nu_2 = I(J-1)$

ANOVA Table

Source of Variation	df	Sum of squares	Mean Square	*f*
Treatments	$I - 1$	SSTr	MSTr	MSTr/MSE
Error	$I(J - 1)$	SSE	MSE	
Total	$IJ - 1$	SST		

Example

Four samples of chicken height of chicken populations in four villages:

i	**Village**	**Height (cm)**					
1	Cikijing	18.7	21.1	17.9	19.5	22.1	18.3
2	Cibejo	19.9	17.6	18.2	20.0	16.9	17.5
3	Cikajang	18.6	20.3	21.7	19.7	20.9	20.8
4	Cibeber	19.1	18.9	18.4	18.8	17.7	20.5

Do ANOVA to determine whether all population means of chicken height are the same or not. Use 5% significance level.

Example

Answer:

i	Village	Height (cm)						Sample mean	Sample variance
1	Cikijing	18.7	21.1	17.9	19.5	22.1	18.3	19.60	2.78
2	Cibejo	19.9	17.6	18.2	20.0	16.9	17.5	18.35	1.71
3	Cikajang	18.6	20.3	21.7	19.7	20.9	20.8	20.33	1.16
4	Cibeber	19.1	18.9	18.4	18.8	17.7	20.5	18.90	0.86

$$I = 4, \;\; J = 6, \;\; \text{df's}: \; \nu_1 = I - 1 = 3, \;\; \nu_2 = I(J-1) = 20 \qquad \bar{x}_{..} = \frac{1}{(4)(6)} \sum_{i=1}^{4} \sum_{j=1}^{6} x_{ij} = 19.296$$

$$\text{MSTr} = \frac{6}{4-1}[(19.60 - 19.296)^2 + (18.35 - 19.296)^2 + (20.33 - 19.296)^2 + (18.90 - 19.296)^2]$$

$$= 4.43$$

$$\text{MSE} = \frac{1}{4}(2.78 + 1.71 + 1.16 + 0.86) = 1.63$$

$$\text{SSTr} = (I-1)\text{MSTr} = (3)(4.43) = 13.29$$

$$\text{SSE} = I(J-1)\text{MSE} = (20)(1.63) = 32.6$$

$$\text{SST} = \text{SSTr} + \text{SSE} = 13.29 + 32.6 = 45.89$$

$$f = \text{MSTr} / \text{MSE} = 4.43 / 1.63 = 2.72$$

$$F_{0.05,3,20} = 3.10 \qquad f < F_{0.05,3,20} \rightarrow \textbf{Accept } H_0$$

Example

ANOVA Table

Source of Variation	df	Sum of squares	Mean Square	*f*
Treatments	3	13.29	4.43	2.72
Error	20	32.6	1.63	
Total	23	45.89		

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